This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

A Bilevel Edge Computing Architecture and Collaborative Offloading Mechanism for Offshore Buoys Network

Jiaxuan Xie, *Member, IEEE*, Linling Kuang, *Member, IEEE*, and Honglei Mu, *Member, IEEE*

***Abstract*—Offshore buoys serve as crucial components and platforms for establishing an initial offshore Internet of Things (IoT) network, enabling future 6G networks. However, the limited size of offshore buoys, and heterogeneous satellite systems, pose a significant challenge for efficient data transmission from offshore buoys to terrestrial gateway. To tackle this challenge, a bilevel edge computing architecture including detection buoys, gateway buoys, and remote cloud is proposed, where detection buoys collect edge information and send the edge computing tasks to gateway buoys (first level) or remote cloud (second level) equipped with computing and storage resources. Then, a collaborative offloading problem is formulated by allocating computing and communication resources based on this bilevel edge computing architecture, which is a typical NP-hard problem and challenging to solve. Thus, a novel distributed algorithm, named hierarchical solution and joint distributed iterative (HSJDI) algorithm, is proposed aimed at reducing computational complexity by decomposition method. Simulation results demonstrate that the proposed HSJDI algorithm performs better than the baseline algorithm, which implies that the total energy consumption is effectively reduced. Besides, we verify our offshore buoys platform and bilevel edge computing architecture by a series of sea trials.**

***Index Terms*—Bilevel edge computing architecture, collaborative offloading, distributed algorithm.**

I. Introduction

THE ocean is not only rich in resources but also holds immense economic significance. Developing an ocean network is a key direction for the future of 6G technology.

Offshore buoys, as vital tools for building 6G networks, play a crucial role in forming an initial offshore buoys Internet of Things (IoT) network, serving as an important platform for developing the maritime economy. We refer to such network of offshore buoys as offshore buoys network (OBN). A significant amount of research has been conducted both domestically and internationally on the OBN. The test platform for OBN began with the ARGO project in 1998. The project envisioned deploying a satellite-tracking buoy every 300 km across the world's oceans to form a vast global ocean observation network. However, the large size and high cost of ARGO buoys have made their widespread deployment difficult. In 2017, the U.S. Defense Advanced Research Projects Agency (DARPA) initiated the "Oceans of Things" project [1]. In 2020, DARPA collaborated with the NATO Centre for Ocean Research and Experimentation to deploy thousands of small, low-cost smart floating devices in the ocean to establish a distributed sensor network. Each smart floating device is equipped with various sensors to collect data such as location information, ocean temperature, salinity, currents, and even activity information of ships, aircraft, and maritime life. These data are transmitted via satellite to the cloud for storage and real-time analysis, providing continuous situational awareness capabilities over vast ocean areas.

However, the existing OBN faces significant challenges. First, the size of the current offshore buoys is extremely limited, only supporting the installation of narrowband satellite communication terminals, which are inadequate for high-speed data backhaul. Second, the communications satellites currently serving OBN are independent and rigidly managed, making it difficult to ensure effective integration between heterogeneous satellite systems. Finally, the transmission rate of offshore buoys range from kilobits per second (kbps) for buoy detection data to Giga bits per second (Gbps) for remote sensing image return data, which increases the complexity of resource management.

To solve such issues, a bilevel edge computing architecture and collaborative offloading mechanism are proposed in this article, which is essential to solve the traditional communication challenges mentioned above by edge computing capability. Specifically, the proposed framework processes data locally at edge nodes in real time, significantly reducing backhaul transmission volume, and effectively mitigating the aforementioned challenges. Then, we divide the buoys into two types of nodes, including gateway buoys and detection buoys, as illustrated in Fig. 1, where detection buoys aim to collect edge information, and such information was sent to gateway buoys equipped with computing and storage resources, which aims to fuse and process edge information. Then, the gateway buoys send the edge fusion information to the remote cloud via the satellite backhaul.

The bilevel edge computing architecture with a three-layer information processing method is proposed, as illustrated in

Received 20 February 2025; revised 5 April 2025; accepted 22 April 2025. *(Corresponding author: Linling Kuang.)*

Jiaxuan Xie and Linling Kuang are with the Department of Electronic Engineering and Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: xiejx19@mails.tsinghua.edu.cn; kll@tsinghua.edu.cn).

Honglei Mu is with the Research Institute of Network Science and Cyberspace, Tsinghua University, Beijing 100084, China (e-mail: mhl23@mails.tsinghua.edu.cn).

Digital Object Identifier 10.1109/TIE.2025.3567704

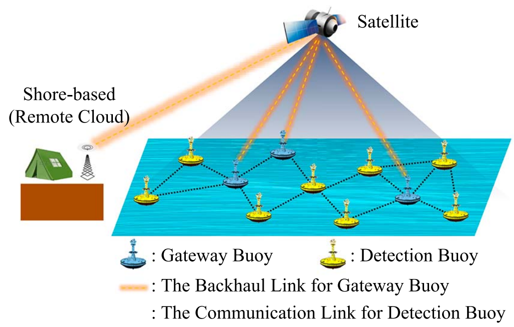


Fig. 1. Typical communication scenario of maritime satellite IoT, including gateway buoys and detection buoys.

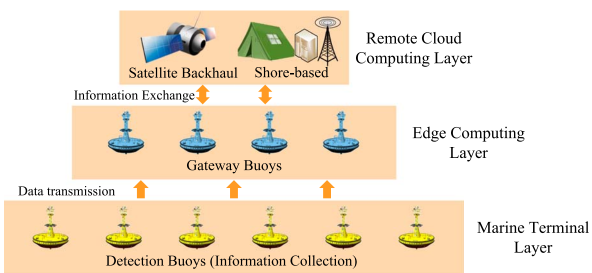


Fig. 2. Bilevel edge computing architecture with remote cloud computing layer and edge computing layer.

Fig. 2. This architecture consists of the marine terminal layer, edge computing layer, and remote cloud computing layer. The marine terminal layer is composed of detection buoys, the edge computing layer consists of gateway buoys, and the remote cloud computing layer includes satellite backhaul and the shore-based command post. Compared with conventional single-layer architectures, our proposed bilevel edge computing architecture demonstrates enhanced adaptability in OBN, effectively coordinating communication resources within the marine terminal layer, thereby substantially reducing the communication overhead within the detection buoys and gateway buoys. In addition, this study intentionally refrains from adopting multilayered edge computing architectures. This is because in terms of device functions, this article only has three layers of network architecture: cloud, detection buoys, and gateway buoys. Further subdivision of detection buoys would increase communication overhead and disperse limited computational resources, ultimately degrading operational efficiency. In this bilevel architecture, the detection gateway exclusively aggregates data to the gateway buoy without direct satellite communication. Marine data transmission is accomplished through buoy gateways equipped with heterogeneous satellite terminals. This design achieves dual objectives: On the one hand, minimizes operational expenses of buoy nodes by centralizing satellite communication interfaces. On the other hand, enables dynamic gateway resource allocation through centralized data aggregation, particularly effective in heterogeneous satellite environments requiring adaptive scheduling.

Nevertheless, several challenges remain to be addressed. Specifically, the issue of limited edge computing resource as well as the collaboration mechanism of cloud-edge are still needed to be discussed. Then, a large body of the existing literature has explored resource collaborative offloading mechanisms to enhance the coordinated scheduling capabilities of edge and cloud computing. The existing literature can be broadly categorized into three main approaches.

The first approach is game theory [2], [3], [4], [5]. A hybrid Stackelberg-auction game approach is proposed in [6], which aims to incent IoT devices to process tasks. Additionally, Lyu et al. [7] proposes a task offloading algorithm based on coalition game theory, which aims to achieve a Nash-stable solution. In [8], an AUV-assisted underwater acoustic covert communication game model is proposed for countermeasures between covert communicators and detectors in ocean IoT, jointly analyzing utility functions, equilibrium strategies, concealment/outage probabilities, and achieving performance improvements over existing methods through Nash equilibrium derivation. However, these studies overlook the fact that maritime IoT devices can be scheduled uniformly due to the fact that all devices are controlled by a unified operations control center, which implies that traditional game theory may not be suitable for the maritime communication scenario.

The second approach is deep learning (DL) [9], [10], [11] or deep reinforcement learning (DRL) [12]. In [13], a comprehensive review of reinforcement learning applications is presented for Internet-of-Underwater-Things challenges, exploring tailored RL algorithms for dynamic underwater communication environments, highlighting performance advantages over conventional and ML-based methods, and outlining open research problems with future directions. In [14], the optimal edge server selection problem for ship IoT users is studied and formulated as a multiarmed bandit problem. Furthermore, Wei et al. [15] proposed a green MEC-enabled maritime IoT network architecture that can flexibly support computing-intensive and latency-sensitive applications. However, DRL algorithms are highly sensitive to training parameters, and the trained neural network can only be used with the current set of parameters, which limits the applicability of DRL algorithms in dynamic environments.

The third approach is classical convex optimization [16], [17], [18]. In [19], a hybrid offshore and aerial-based multiaccess edge computing scheme is proposed for marine communication networks and solved by a joint optimization problem. Additionally, Yang et al. [20] proposed a two-stage joint optimization algorithm to optimize computation and communication resources under delay and energy constraints. In [21], a time-competition forwarding strategy is proposed to prevent redundant forwarding data copies from communication as much as possible in the Internet of Marine Things (IoMaT). In [22], a hybrid LEO and UAV edge computing method is proposed for marine IoT systems in space-air-sea integrated networks, jointly optimizing bit allocation and UAV trajectory planning under latency and energy constraints using successive convex approximation (SCA) strategies for different LEO accessibility cases. While classical convex optimization can achieve better performance without the need for pretraining neural network parameters, it involves high algorithmic complexity and cannot support real-time calculation and task allocation. Notably, energy efficiency constitutes the primary design consideration for maritime IoT systems rather than latency optimization. We therefore prioritize energy-efficient system design in our framework, with particular emphasis on achieving minimal energy consumption through resource allocation strategies.

To address the aforementioned challenges, this article proposes a distributed, low-complexity optimization method to achieve fast offloading of computing tasks and real-time resource allocation. The main contributions are summarized as follows.

1) A typical communication scenario of maritime IoT is analyzed, and a novel bilevel edge computing architecture is proposed. Then, a collaborative offloading problem is formulated with the objective of minimizing the total energy consumption. This problem, characterized by first- and second-level offloading constraints, is a typical NP-hard problem and challenging to solve.
2) To address this collaborative offloading problem, a novel hierarchical solution and joint distributed iterative (HSJDI) algorithm is proposed. Specifically, the algorithm first determines offloading variables, followed by an iterative process to optimize the communications resources variables and computing resources variables. It incorporates dual ascent (DA), SCA, block coordinate descent (BCD), and dual decomposition techniques, providing a strong theoretical foundation for its performance.
3) Numerical results illustrate that our HSJDI algorithm performs better than the baseline algorithm, and the total energy consumption is effectively reduced. In addition, we verify our offshore buoys and bilevel edge computing architecture by hardware platform.

The rest of this article is organized as follows. Section II describes the system model. In Section III, a novel HSJDI algorithm is proposed. Numerical simulation results are provided in Section IV, and conclusions are given in Section V.

II. System Model

Let $\mathcal{N}=\{1,2,\ldots,N\}$ denotes the set of offshore buoys, including gateway buoys $\mathcal{N}^M=\{1,2,\ldots,M\}$ and detection buoys $\mathcal{N}^Q=\{1,2,\ldots,Q\}$, which implies that $N=M+Q$. While detection buoy aims to monitor maritime conditions and transmit the sensed data back to the gateway buoy for fusion calculation and processing. Gateway buoy, equipped with edge computing resource, processes the data from the detection buoy and transmits the processed results to the remote cloud.

A. Computing Model

A two-tuple $\Psi_q=\{I_q,\phi_q\}$ is proposed to depict the transmission data I_q and calculation data ϕ_q for the detection buoy q. Define the offloading variables from the detection buoy to the gateway buoy as a binary matrix $\mathcal{X}$ with the size of $Q\times M$, where $x_{qm}=1$ means that the computing task generated by detection buoy q is offloaded to the gateway buoy m. $x_{qm}=0$, otherwise. Let $\mathcal{C}$ denotes the CPU resource allocation matrix of size $Q\times M$, where c_{qm} represents the CPU resources allocated from gateway buoy m to detection buoy q. Let $\mathcal{Y}$ denotes the set of offloading variables for gateway buoys with the size of $Q\times M$, where $y_{qm}\in[0,1]$ represents the ratio of gateway buoy m offloaded to remote cloud.

1) Edge Computing Delay: Therefore, the edge computing time T_m^{s1} of gateway buoy m can be given as follows [23]:

$$T_m^{s1}=\sum_{q=1}^{Q}T_{qm}^{s1}=\sum_{q=1}^{Q}\frac{x_{qm}\phi_q(1-y_{qm})}{c_{qm}} \tag{1}$$

where T_{qm}^{s1} denotes edge computing time for gateway buoy m when computing the task of detection buoy q. While $x_{qm}=0$ means detection buoy q does not offload its task to gateway buoy m, which causes gateway buoy m does not choose.

2) Edge Computing Energy Consumption: The edge computing energy consumption E_m^s of gateway buoy $m\in\mathcal{N}^M$ should be considered due to limited edge energy, which is illustrated as follows according to the literature [19]:

$$E_m^s=\epsilon_m\sum_{q=1}^{Q}c_{qm}^3T_{qm}^{s1}=\epsilon_m\sum_{q=1}^{Q}c_{qm}^2x_{qm}\phi_q(1-y_{qm}) \tag{2}$$

where ϵ_m is the power consumption coefficient of edge gateway buoy m, and edge computing energy consumption scales cubically with CPU utilization multiplied by runtime duration [24]. Then, by substituting into (1), formula (2) is established.

B. Communication Model

Let $\mathcal{W}=\{\mathcal{N},\mathcal{L},\mathcal{R}\}$ denotes the OBN, where $\mathcal{N}$ denotes the set of offshore buoys. $\mathcal{L}$ is the set of communication links, which is denoted as a connection matrix of offshore buoys with the size of $N\times N$, where $l_{i,j}=1$ represents buoy i is directly connected to buoy j, and $l_{i,j}=0$ otherwise. In addition, $\mathcal{R}$ will be introduced in what follows.

1) Maritime Channel Model: The fading models of maritime communication channel can be classified into buoy-to-buoy channel and buoy-to-satellite channel. Specifically, buoy-to-buoy channel consists of the direct path on the sea and the reflection path on the sea surface. Then, according to [25] [(2)–(6)], the channel coefficient among buoys i,j can be modeled as

$$h_{ij}^{\mathrm{IoT}}=G_0\cdot 2\sin\left(\frac{2\pi h_ih_j}{\lambda_1 d_{ij}^{\mathrm{IoT}}}\right)\cdot\frac{\lambda}{4\pi d_{ij}^{\mathrm{IoT}}} \tag{3}$$

where λ denotes the wavelength, d_{ij}^{IoT} represents the distance between the transmitting and receiving antennas, and G_0 denotes the antenna gain of buoys, while h_i and h_j correspond to the heights of the transmitting and receiving antennas, respectively. In addition, buoy-to-satellite channel is mainly a LoS path, and the channel coefficient is calculated as follows:

$$h_i^{\mathrm{Sat}}=\sqrt{G_iG_{\mathrm{Sat}}}\cdot\frac{\lambda_2}{4\pi d_i^{\mathrm{Sat}}} \tag{4}$$

where G_i and G_{Sat} are transmitting antenna gain of buoy and receiving antenna gain of satellite. λ_2 denotes the wavelength, and d_i^{Sat} is the distance between the buoy and satellite.

2) Transmission Rate Within Offshore Buoys: The transmission rate between each pair of offshore buoys equipped with directional transmission antennas is calculated as

$$R_{i\to j}(p_{ij}) = B_0 \log_2\left(1+\frac{p_{ij}|h_{ij}^{\mathrm{IoT}}|^2}{\sigma_j^2}\right) \tag{5}$$

where B_0 denotes the transmission bandwidth. Let $\mathcal{P}^b = \{p_{ij}\}_{i\in\mathcal{N},j\in\mathcal{N}}$ denotes the set of transmission power among all buoys equipped with directional antennas, where p_{ij} represents the transmission power from buoy $i\in\mathcal{N}$ to buoy $j\in\mathcal{N}$.

Define the set of relay buoys from detection buoy q to gateway buoy m as follows, when the detection buoy does not have a direct link to gateway buoy:

$$\Gamma_{qm} = \{q, \xi_1, \xi_2, \ldots, \xi_{K_{qm}}, m\} \tag{6}$$

then we have $\xi_k \in \mathcal{N}$, $\Gamma_{qm}\subset\mathcal{N}$, $l_{\xi_k\xi_{k+1}} = 1, \forall k$. The data transmission passed through K_{qm} buoys from source buoy q to sink buoy m. If the buoy $n\in\mathcal{N}$ is included in the path Γ_{qm} (i.e., $n\in\Gamma_{qm}$), then the cumulative number of buoy n appearing on all relay paths is calculated as $u_n = \sum_{m\in\mathcal{N}^M}\sum_{q\in\mathcal{N}^Q}\mathbb{I}_{\{n\in\Gamma_{qm}\}}$. Let $\tilde{\Gamma} = \{\Gamma_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$ denotes the set of all relay buoys path. Define the minimum transmission rate of the relay link from source buoy q to sink buoy m as

$$r_{q\to m} = \min\{R_{q\to\xi_1}, R_{\xi_1\to\xi_2}, \ldots, R_{\xi_{K_{qm}}\to m}\} \tag{7}$$

where $q\in\mathcal{N}^Q$ is detection buoy, and $m\in\mathcal{N}^M$ is gateway buoy. We assume that each buoy can work as a relay node. Then, for each pair of buoys, we define its reachability matrix $\mathcal{R}$ with the size of $Q\times M$, expressed as follows:

$$\mathcal{R} = \begin{bmatrix} r_{1\to1} & r_{1\to2} & \cdots & r_{1\to M}\\ r_{2\to1} & r_{2\to2} & \cdots & r_{2\to M}\\ \vdots & \vdots & \ddots & \vdots\\ r_{Q\to1} & r_{Q\to2} & \cdots & r_{Q\to M}\end{bmatrix} \tag{8}$$

where $r_{q\to m}\in\mathcal{R}$ denotes the communication rate between detection buoy q and gateway buoy m via either a direct connection or relay communication, as calculated by (7).

3) Transmission Rate Between Gateway Buoys and Satellite: Let $\Omega = \{1,2,\ldots,W\}$ represent the set of satellite subcarriers with OFDM communication system (i.e., Starlink [26]). Let Ω_m denote the set of satellite subcarriers allocated to gateway buoy m, it concludes that $\cup_{m=1}^{M}\Omega_m = \Omega$ and $\sum_{m=1}^{M}|\Omega_m| = W$, where $|\Omega_m|$ denotes the number of elements in the set Ω_m. Then, based on (4), the transmission rate between gateway buoy m and satellite is calculated as

$$\begin{aligned} &R_{m\to\mathrm{sat}}(\Omega_m, \{p_{ms}^{\omega}\}_{\omega\in\Omega_m})\\ &\quad= \sum_{\omega\in\Omega_m}\frac{B^{\mathrm{sat}}}{W}\log_2\left(1+\frac{p_{ms}^{\omega}|h_i^{\mathrm{Sat}}|^2}{\sigma_0^2}\right) \end{aligned} \tag{9}$$

where B^{sat} represents the subcarrier Ka-band (or Ku-band) bandwidth in OFDM communication system, σ_0^2 represents the Gaussian white noise, and $p_{m,s}^{\omega}$ denotes the transmitting power from the gateway buoy m to the satellite. To unify the representation, define the set of transmission power p_{ms} as the matrix $\mathcal{P}^s = \{p_{ms}^{\omega}\}_{m\in\mathcal{N}^M,\omega\in\Omega_m}$ in what follows.

4) Relay Transmission Delay: The transmission delay T_{qm}^{c1} from detection buoy q to gateway buoy m is calculated as

$$T_{qm}^{c1} = \begin{cases} \dfrac{I_q}{R_{q\to\xi_1}(p_{q\xi_1})} + \displaystyle\sum_{\xi_k\in\Gamma_{qm}\setminus\{q,m\}}\frac{I_q}{R_{\xi_k\to\xi_{k+1}}(p_{\xi_k\xi_{k+1}})} \\ \quad + \dfrac{I_q}{R_{\xi_{K_{qm}}\to m}(p_{\xi_{K_{qm}}m})}, \quad \text{if } |\Gamma_{qm}|>2 \\ \dfrac{I_q}{R_{q\to m}(p_{qm})}, \ \text{if } |\Gamma_{qm}|=2 \\ 0, \ \text{if} |\Gamma_{qm}|=0 \end{cases} \tag{10}$$

where transmission delay T_{qm}^{c1} is not inherently a constant. It is intrinsically linked to the decision variables $\mathcal{P}^b = \{p_{ij}\}_{i\in\mathcal{N},j\in\mathcal{N}}$, which represent the power allocation strategy. $|\Gamma_{qm}|=2$ denotes $\Gamma_{qm}=\{q,m\}$, which implies that detection buoy q and gateway buoy m can communicate directly. While $|\Gamma_{qm}|>2$ denotes $\Gamma_{qm}=\{q,\xi_1,\xi_2,\ldots,\xi_{K_{qm}},m\}$, which implies that $x_{qm}=1$ holds, and the data generated by detection buoy q needs to be relayed by other buoys to reach the gateway buoy m. $|\Gamma_{qm}|=0$ denotes $\Gamma_{qm}=\emptyset$, which means that $x_{qm}=0$ holds, that is, the data generated by detection buoy q will not be offloaded to the gateway buoy m. In addition, $R_{\xi_k\to\xi_{k+1}}(p_{\xi_k\xi_{k+1}})$ is calculated by (5). It is evident that the first-level offloading decision x_{qm} can affect the result of transmission delay T_{qm}^{c1}.

5) Backhaul Transmission Delay: The backhaul transmission delay of all gateway buoys denotes the transmission delay between gateway buoys and satellite, which is calculated as

$$T_{ms}^{c2} = \frac{\sum_{q=1}^{Q} x_{qm}(\epsilon_q I_q) y_{qm}}{\sum_{\omega\in\Omega_m}\frac{B^{\mathrm{sat}}}{W}\log_2\left(1+\frac{p_{ms}^{\omega}|h^{\mathrm{Sat}}|^2}{\sigma_0^2}\right)} \tag{11}$$

where the numerator represents the total backhaul data for gateway buoy m, and ϵ_q is a coefficient of I_q.

6) Communication Energy Consumption: The communication energy consumption E_n^c of offshore buoy $n\in\mathcal{N}$ can be divided into detection buoy communication energy consumption E_m^c and gateway buoy energy consumption E_q^c.[1] E_m^c includes transmission energy consumption within offshore buoys E_m^{c1} and transmission energy consumption between gateway buoys and satellite E_m^{c2}, where E_m^c is calculated as

$$\begin{aligned} E_m^{c1} &= \sum_{q=1}^{Q} E_{mq}^{c1} = \sum_{q=1}^{Q}\mathbb{I}_{\{m\in\Gamma_{q\tilde{m}}\}}\cdot T_{m\tilde{\xi}^q}^{c1}\cdot p_{m\tilde{\xi}^q}\\ &= \sum_{q=1}^{Q}\frac{\mathbb{I}_{\{m\in\Gamma_{q\tilde{m}}\}} I_q\cdot p_{m\tilde{\xi}^q}}{B_0\log_2\left(1+\frac{p_{m\tilde{\xi}^q}|h_{m\tilde{\xi}^q}^{\mathrm{IoT}}|^2}{\sigma_{\tilde{\xi}^q}^2}\right)} \end{aligned} \tag{12}$$

where E_m^{c1} implies that the gateway buoy m plays the role of a relay buoy, when the data were sent from source buoy q to sink buoy $\tilde{m}$ (i.e., $m\in\Gamma_{q\tilde{m}}$), and the next hop of the relay link is buoy $\tilde{\xi}^q$ (i.e., $l_{m\tilde{\xi}^q}=1$, $\tilde{\xi}^q\in\Gamma_{q\tilde{m}}$). E_q^{c1} can be provided in a manner similar to E_m^{c1}. In addition, it is worth mentioning that if the gateway buoy m is not used as a relay node, then $m\notin\tilde{\Gamma}$, which

[1]Since the energy consumption calculation formulas of gateway buoys and detection buoys are different, we have to use different symbols (i.e., m and q) to distinguish them which aims to avoid confusion.

implies that $E_m^{c1} = 0$ holds in (12). In addition, E_m^{c2} is calculated as follows:

$$E_m^{c2} = \sum_{\omega\in\Omega_M} E_{m\omega}^{c2} = \sum_{\omega\in\Omega_M}\left(\frac{\sum_{q=1}^{Q} x_{qm} I_q y_{qm}}{\frac{B^{\text{sat}}}{W}\log_2\left(1+\frac{p_{ms}^{\omega}|h^{\text{Sat}}|^2}{\sigma_0^2}\right)}\cdot p_{ms}^{\omega}\right) \tag{13}$$

where E_m^{c2} measures the energy consumption of transmission from the gateway buoy m to the satellite according to the formulas (9) and (11). Then, the total communication energy consumption of gateway buoy m is calculated as follows:

$$E_m^c = \epsilon_c^1 \cdot E_m^{c1} + \epsilon_c^2 \cdot E_m^{c2} \tag{14}$$

where E_m^{c1} and E_m^{c2} are illustrated as (12) and (13), ϵ_c^1 and ϵ_c^2 are proportional coefficients. Besides, for communication energy consumption of detection buoy q, we conclude that E_q^c only equals to E_q^{c1}, and the formula is similar to (12), which is omitted for brevity.

C. Collaborative Offloading Problem

Based on the above description, a collaborative offloading problem is proposed, which aims to minimize the total energy consumption of joint computing and communication system

$$E^{\text{Total}} = \sum_{m=1}^{M} E_m^c + \sum_{i=1}^{N} E_i^c + \sum_{i=1}^{N} E_i^0 \tag{15}$$

where $\sum_{m=1}^{M} E_m^c$ is the computing energy consumption, and $\sum_{i=1}^{N} E_i^c$ is the communication energy consumption. $\sum_{i=1}^{N} E_i^0$ denotes the inherent baseline energy consumption required for system startup and sustained operation. Then, the detail optimization problem is illustrated as follows:

$$\boldsymbol{P_1}: \min_{\mathcal{X},\mathcal{Y},\mathcal{C},\mathcal{P}^s,\mathcal{P}^b,\Omega} E^{\text{Total}} \tag{16a}$$

$$\text{s.t.} \begin{cases} \text{First-Level Offloading Constraints}: \\ \quad C_1: \sum_{q=1}^{Q} x_{qm} = 1, x_{qm}\in\{0,1\}, \forall m\in\mathcal{N}^M \\ \quad C_2: \sum_{q=1}^{Q} c_{qm} \le C_0, \forall m\in\mathcal{N}^M \\ \quad C_3: \sum_{n=1}^{N} p_{qn} \le P_0, \forall q\in\mathcal{N}^Q \\ \quad C_4: r_{q\to m} \ge R_{\min}, \forall q\in\mathcal{N}^Q \\ \text{Second-Level Offloading Constraints}: \\ \quad C_5: 0\le y_{qm}\le 1, \forall q\in\mathcal{N}^Q, \forall m\in\mathcal{N}^M \\ \quad C_6: \sum_{n=1}^{N} p_{mn} + \sum_{\omega\in\Omega_M} p_{ms}^{\omega} \le P_0, \forall m\in\mathcal{N}^M \\ \quad C_7: R_{m\to sat}(\Omega_m, \{p_{ms}^{\omega}\}_{\omega\in\Omega_m}) \ge R_{\min}, \forall m \end{cases} \tag{16b}$$

where the decision variables $\mathcal{X} = \{x_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$ are the first-level offloading decisions (binary matrices) between detection buoys and gateway buoys. The decision variables $\mathcal{Y} = \{y_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$ are the second-level offloading decisions between gateway buoys and remote cloud. The decision variables $\mathcal{C} = \{c_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$ are the CPU resource allocation matrices of gateway buoy m for detection buoy q's tasks. The decision variables $\mathcal{P}^s = \{p_{ms}^{\omega}\}_{m\in\mathcal{N}^M, \omega\in\Omega_m}$ are the transmission power allocations between gateway buoys and satellites. The decision variables $\mathcal{P}^b = \{p_{ij}\}_{i\in\mathcal{N}, j\in\mathcal{N}}$ are the transmission power levels within offshore buoys. The decision variable $\Omega = \{1,2,\ldots,W\}$ is the set of satellite subcarriers. Then, the constraints can be divided into two parts: first-level offloading constraints, which aim to restrict the variables within the offshore buoys, and second-level offloading constraints, which focus on the variables between the gateway buoys and detection buoys. Specifically, C_1 constraint implies that each detection buoy with a computing task must select one gateway buoy and offload its data to that gateway buoy. C_2 constraint restricts the total computing resources for each gateway buoy. C_3 and C_6 constraints limit the total transmission power of each detection buoy and gateway buoy, respectively. C_4 and C_7 constraints define the minimum transmission rate for computing tasks.

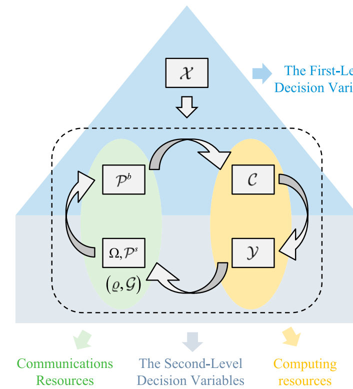


Fig. 3. HSJDI algorithm first solves $\mathcal{X}$, followed by an iterative process to determine the variables $\mathcal{Y}$, $\mathcal{C}$, $\mathcal{P}^b$, Ω, and $\mathcal{P}^s$.

III. Problem Solving

It is worth mentioning that the collaborative offloading problem P_1, regarded as a mixed integer programming (MIP) problem [27], is difficult to solve. A novel HSJDI algorithm aimed at reducing computational complexity by decomposition method is proposed as illustrated in Fig. 3. Specifically, decision variables $\mathcal{X}, \mathcal{P}^b, \mathcal{C}$ are the first-level variables between marine terminal layer and edge computing layer. Decision variables $\mathcal{P}^s, \Omega, \mathcal{Y}$ are the second-level variables between edge computing layer and remote cloud computing layer.

A. Problem Transformation and Approximation

1) Problem Transformation: This section introduces the transformation method for P_1. $r_{q\to m}$ in C_4 constraint is the minimum value of multiple transmission rate according to (7), which implies that, for $\forall q$, C_4 constraint is rewritten as

$$C_4': \begin{cases} R_{q\to\xi_1} \ge R_{\min} \\ R_{\xi_k\to\xi_{k+1}} \ge R_{\min}, k\in\{1,2,\ldots,K_{qm}-1\} \\ R_{\xi_{K_{qm}}\to m} \ge R_{\min} \end{cases} \tag{17}$$

Algorithm 1: Solving for First-Level Offloading Decision $\{\mathcal{X}\}$.

Input: Optimization Problem P_1.
Output: $\{\mathcal{X}\} = \{x_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$.
1: **Initialize** For $\forall q$, select the maximum communication rate $\max_{m\in\mathcal{N}^M} r_{q\rightarrow m}$ as initial connection $\{\hat{x}^d_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$.
2: **while** (1) **do**
3: Calculate task load for each gateway buoy m based on $\hat{c}^d_m = \sum_{q=1}^{Q} \hat{x}^d_{qm} \cdot I_q$.
4: Normalize $\{\hat{c}^d_m\}_{m\in\mathcal{N}^M}$ and output $\{\tilde{c}^d_m\}_{m\in\mathcal{N}^M}$, where $\tilde{c}^d_m \in [0, 1]$.
5: Calculate the variance of $\{\tilde{c}^d_m\}_{m\in\mathcal{N}^M}$, and obtain σ^2.
6: **if** $\sigma^2 < \sigma^2_0 \vee k > K_0$ **then**
7: **BreakWhile**
8: **end if**
9: Update $k = k + 1$, select K_1 gateway buoys, and reallocate computing tasks based on heuristic methods.
10: **end while**
11: **return** $\boldsymbol{\mathcal{X}}$

where gateway buoy m satisfies $x_{qm} = 1$, and $\xi_k \in \Gamma_{qm}$. According to (5), it concludes that C'_4 constraint is a negative logarithmic function of $\mathcal{P}^b$, which is a convex constraint.

In addition, (9) includes integer variables Ω and continuous variables $\mathcal{P}^s$, which are difficult to solve. Then, a classic time-sharing condition [28] is proposed to solve this issue based on the sharing factor $\rho^{\omega}_m \in [0, 1]$. The physical meaning of ρ^{ω}_m is the proportion of time-slot that subcarrier $\omega \in \Omega$ is assigned to gateway buoy m. Define $g^{\omega}_{ms} = \rho^{\omega}_m \cdot p^{\omega}_{ms}$ as the actual transmit power. Let $\varrho = \{\rho^{\omega}_m\}_{m\in\mathcal{N}^M,\omega\in\Omega}$ and $\mathcal{G} = \{g^{\omega}_{ms}\}_{m\in\mathcal{N}^M,\omega\in\Omega}$ denote the set of sharing factors and actual transmit powers, respectively. Then, the transmission rate between gateway buoy m and satellite is calculated as

$$R_{m\rightarrow \mathrm{sat}}\left(\{\rho^{\omega}_m\}_{\omega\in\Omega_m}, \{g^{\omega}_{ms}\}_{\omega\in\Omega_m}\right) = \sum_{\omega=1}^{W} \rho^{\omega}_m \left[\frac{B^{\mathrm{Sat}}}{W} \log_2\left(1 + \frac{g^{\omega}_{ms}|h^{\mathrm{Sat}}|^2}{\rho^{\omega}_m \cdot \sigma^2_0}\right)\right] \quad (18)$$

then the formulas of backhaul transmission delay T^{c2}_{ms} in (11), and transmission energy consumption between gateway buoys and satellite E^{c2}_m in (13), can be rewritten based on (18).

2) Solving for First-Level Offloading Decision: This section solves the first-level offloading decision $\mathcal{X} = \{x_{qm}\}_{q\in\mathcal{N}^Q, m\in\mathcal{N}^M}$, which aims to determine the mapping relationship between detection buoys and gateway buoys. It is worth mentioning that this suboptimization problem is an NP-hard problem with deeply coupled variables. Therefore, a heuristic strategy is proposed in Algorithm 1, which conforms to the intuitive solution strategy. Specifically, line 1 provides an intuitive initial allocation based on the transmission rate $r_{q\rightarrow m}$ (or distance d^{IoT}_{ij}). Let k denotes the iteration index. Then, the traffic load of each gateway node is normalized. Subsequently, nodes with heavy loads are identified, and heuristics or rules are applied to redistribute tasks accordingly. For brevity, the specific heuristic rules are not detailed here.

B. Joint Iterative Mechanism

A joint iterative mechanism based on the classic BCD method is proposed for solving $\{\varrho, \mathcal{G}\}$, $\mathcal{P}^b$, $\mathcal{C}$, and $\mathcal{Y}$ sequentially based on Fig. 3. Due to limited space, only the solution of variable $\{\varrho, \mathcal{G}\}$ is given, and the solution of other variables can be found in [29].

1) Solving for Decision Variables $\{\varrho, \mathcal{G}\}$: In this section, the derivation of the decision variables ϱ and $\mathcal{G}$ is introduced, when other decision variables are fixed, and the optimization problem P^{Sub}_2 is given as follows:

$$\boldsymbol{P^{\mathrm{Sub}}_2} : \min_{\varrho,\mathcal{G}} \sum_{m=1}^{M}\sum_{\omega=1}^{W} E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms}) \quad (19a)$$

$$\text{s.t.} \quad C_6, C_7 \quad (19b)$$

where the objective function is illustrated as follows based on the time-sharing conditions:

$$E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms}) = \frac{\sum_{q=1}^{Q} x_{qm} I_q y_{qm} \cdot g^{\omega}_{ms}}{\rho^{\omega}_m \left[\frac{B^{\mathrm{Sat}}}{W}\log_2\left(1 + \frac{g^{\omega}_{ms}|h^{\mathrm{Sat}}|^2}{\rho^{\omega}_m\cdot\sigma^2_0}\right)\right]} \quad (20)$$

where $\sum_{q=1}^{Q} x_{qm} I_q y_{qm}$ can be regarded as a constant.

First, $E^{c2}_{m\omega}$ is a nonconvex function with respect to decision variables $\{\rho^{\omega}_m, g^{\omega}_{ms}\}$. To solve such problem, a novel iterative algorithm, based on SCA method, is proposed to approximate this nonconvex objective function. Specifically, the nonconvex constraint $E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms})$ can be regarded as the product of two convex functions

$$E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms}) = (c_1 \cdot g^{\omega}_{ms}) \cdot \left[\frac{1}{\rho^{\omega}_m \log_2\left(1 + \frac{g^{\omega}_{ms}|h^{\mathrm{Sat}}|^2}{\rho^{\omega}_m\cdot\sigma^2_0}\right)}\right] \quad (21)$$

where $c_1 = (W \cdot \sum_{q=1}^{Q} x_{qm} I_q y_{qm} / B^{\mathrm{Sat}})$ is a constant. We will analyze the convexity of $(1/\rho^{\omega}_m \log_2(1 + (g^{\omega}_{ms}|h^{\mathrm{Sat}}|^2/\rho^{\omega}_m \cdot \sigma^2_0)))$ with the function form of $f(x, y) = (1/x \log_2(1 + (y/x)))$. According to [30], a 2-D function is convex if its projection onto any line remains convex. Thus, for any $a \in \mathbb{R}$ and $b \in \mathbb{R}$, let $y = ax + b$. In this case, we determine the convexity of $f(x, y)$ by examining $h(x)$ instead

$$h(x) = f(x,y)|_{y=ax+b} \overset{(a)}{=} \frac{1}{x\log_2(1 + a + \frac{b}{x})} \overset{(b)}{\underset{\triangle}{=}} \frac{1}{g(x)} \quad (22)$$

where substituting the formula of function $f(x, y)$, we can see that equation (a) holds. Equation (b) is the definition of $g(x)$. Then, the second-order derivative of function $g(x)$ is calculated as $g''(x) = -(b^2 \ln 2/x((1 + a)x + b)^2)$, where $g''(x) > 0$ holds when $x > 0$, which implies $g(x)$ is a concave function. Then, we can analyze the convexity of the function $h(x)$ based on the concavity of $g(x)$. Note that $h(x)$ can be regarded as the inverse proportional function of the $g(x)$ function. In addition, the outer function (i.e., inverse proportional function) is a convex function and nonincreasing, and the inner function (i.e., $g(x)$) is a concave function. According to the composition rules [31], $h(x)$ is a convex function, which implies that $f(x, y)$ is a convex function, and the objective function $E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms})$ can be regarded as the product of two convex functions. Next, the convex upper bound $E^{c2}_{m\omega}(\rho^{\omega}_m, g^{\omega}_{ms}; \tilde{\rho}^{\omega}_m, \tilde{g}^{\omega}_{ms})$ for nonconvex function $E^{c2}_{m\omega}(t^{\omega}_m)$ is derived in (23), shown at the bottom of the page, based on SCA, where $G(g_{ms}^{\omega}, \rho_m^{\omega}) = (1/2)\cdot(c_1\cdot g_{ms}^{\omega} + (\rho_m^{\omega}\log_2(1+(g_{ms}^{\omega}|h^{\text{Sat}}|^2/\rho_m^{\omega}\cdot\sigma_0^2)))^{-1})^2$ and $H(\rho_m^{\omega}, g_{ms}^{\omega}) = -(1/2)\cdot(\rho_m^{\omega}\log_2(1+(g_{ms}^{\omega}|h^{\text{Sat}}|^2/\rho_m^{\omega}\cdot\sigma_0^2)))^{-2}$ hold.

Second, let us discuss the convexity of C_6 constraint, as mentioned in (19b), which can be rewritten as follows based on the time-sharing conditions:

$$\begin{aligned} C_6' &: \sum_{n=1}^{N} p_{mn} + \sum_{\omega=1}^{W} g_{ms}^{\omega} \le P_0, \forall m \in \mathcal{N}^M \\ &\Rightarrow \sum_{\omega=1}^{W} g_{ms}^{\omega} + c_2 \le 0, \forall m \in \mathcal{N}^M \end{aligned} \tag{24}$$

where $c_2 = \sum_{n=1}^{N} p_{mn} - P_0$ is a constant when solving decision variables $\{\varrho, \mathcal{G}\}$. Note that C_6' is a linear constraint, which implies that C_6' constraint is a convex constraint.

Thirdly, the analysis of C_7 constraint is illustrated as

$$C_7' : -\sum_{\omega=1}^{W} \rho_m^{\omega}\left[\frac{B^{\text{Sat}}}{W}\log_2\left(1+\frac{g_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)\right] + R_{\min} \le 0, \forall m$$

where it concludes that $g(x)$ is a concave function, which implies that $-g(x)$ is a convex function. Thus, C_8' constraint is a convex constraint.

Next, based on the above analysis, it concludes P_2^{Sub} in (19) is a convex optimization problem. Therefore, a classical DA [32] method can be used to solve this problem. Then, the Lagrangian function of P_2^{Sub} is expressed as

$$\begin{aligned} &L_2\left(\{\varrho, \mathcal{G}\}, \{\alpha_m, \beta_m, \eta_m\}_{m\in\mathcal{N}^M}\right) \\ &= \sum_{m=1}^{M}\sum_{\omega=1}^{W} \tilde{E}_{m\omega}^{c2}\left(\rho_m^{\omega}, g_{ms}^{\omega}; \tilde{\rho}_m^{\omega}, \tilde{g}_{ms}^{\omega}\right) + \sum_{m=1}^{M}\alpha_m\left(\sum_{\omega=1}^{W} g_{ms}^{\omega} + c_2\right) \\ &+ \sum_{m=1}^{M}\beta_m\left[-\sum_{\omega=1}^{W}\rho_m^{\omega}\left[\frac{B^{\text{Sat}}}{W}\log_2\left(1+\frac{g_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)\right] + R_{\min}\right] \\ &\triangleq \sum_{m=1}^{M} L_2^m\left(\{\rho_m^{\omega}, g_{ms}^{\omega}\}_{m\in\mathcal{N}^M}\right) + c_3 \end{aligned} \tag{25}$$

where $L_2^m\left(\{\rho_m^{\omega}, g_{ms}^{\omega}\}_{m\in\mathcal{N}^M}\right)$ is related to m, c_3 is a constant.

Then, the update of decision variable $\varrho = \{\rho_m^{\omega}\}_{m\in\mathcal{M},\omega\in\Omega}$ for the κ th iteration is given as

$$\rho_{ms}^{\omega(\kappa+1)} = \rho_{ms}^{\omega(\kappa)} - \zeta_{mg}^{\omega(\kappa)}\cdot\frac{\partial L_2^m\left(\{\rho_m^{\omega(\kappa)}, g_{ms}^{\omega(\kappa)}\}_{m\in\mathcal{N}^M}\right)}{\partial g_{ms}^{\omega(\kappa)}}. \tag{26}$$

Then, the update of decision variable $\mathcal{G} = \{g_{ms}^{\omega}\}_{m\in\mathcal{M},\omega\in\Omega}$ for the κth iteration is given as

$$g_{ms}^{\omega(\kappa+1)} = g_{ms}^{\omega(\kappa)} - \zeta_{mg}^{\omega(\kappa)}\cdot\frac{\partial L_2^m\left(\{\rho_m^{\omega(\kappa)}, g_{ms}^{\omega(\kappa)}\}_{m\in\mathcal{N}^M}\right)}{\partial g_{ms}^{\omega(\kappa)}} \tag{27}$$

where $\zeta_{mg}^{\omega(\kappa+1)}$ is the update step size for the ith iteration.

Let κ represents the index of DA algorithm for outer loop and $\tilde{\kappa}$ represents the index of SCA algorithm for inner loop. Then, the update of $\tilde{\rho}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}$ and $\tilde{g}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}$ for $(\tilde{\kappa}+1)$th iteration is given as follows:

$$\begin{aligned} &\left\{\tilde{\rho}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}, \tilde{g}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}\right\} \\ &\qquad = \arg\min_{\varrho,\mathcal{G}} \sum_{m=1}^{M}\sum_{\omega=1}^{W}\tilde{E}_{m\omega}^{c2}\left(\rho_m^{\omega}, g_{ms}^{\omega}; \tilde{\rho}_m^{\omega}, \tilde{g}_{ms}^{\omega}\right) \end{aligned} \tag{28a}$$

$$\text{s.t.}\quad C_6', C_7' \tag{28b}$$

where the above formula can be used to update the auxiliary variables in SCA.

Then, the Lagrange multipliers are calculated as

$$\begin{aligned} \alpha_m^{(\kappa+1)} &= \alpha_m^{(\kappa)} + \zeta_{m\alpha}^{(\kappa)}\cdot\left(\sum_{\omega=1}^{W} g_{ms}^{\omega(\kappa)} + c_1\right) \\ \beta_m^{(\kappa+1)} &= \beta_m^{(\kappa)} + \zeta_{m\beta}^{(\kappa)}\cdot\left(-\upsilon\left(\rho_m^{\omega(\kappa)}, g_{ms}^{\omega(\kappa)}\right) + R_{\min}\right) \end{aligned} \tag{29}$$

where $\zeta_{m\alpha}^{(\kappa)}, \zeta_{m\beta}^{(\kappa)}$ are the step sizes for Lagrange multipliers. The solution algorithm for decision variables $\{\varrho, \mathcal{G}\}$ is given in Algorithm 2.

Finally, due to limited space, the other four sections includes solving for decision variables $\{\mathcal{P}^b\}$, solving for decision variables $\{\mathcal{C}\}$, solving for decision variables $\{\mathcal{Y}\}$, and joint iteration

$$\begin{aligned} E_{m\omega}^{c2}(\rho_m^{\omega}, g_{ms}^{\omega}) &= (c_1\cdot g_{ms}^{\omega})\cdot\left[\frac{1}{\rho_m^{\omega}\log_2\left(1+\frac{g_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)}\right] \\ &= \frac{1}{2}\left(c_1\cdot g_{ms}^{\omega} + \frac{1}{\rho_m^{\omega}\log_2\left(1+\frac{g_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)}\right)^2 - \frac{1}{2}(c_1\cdot g_{ms}^{\omega})^2 - \frac{1}{2}\left[\rho_m^{\omega}\log_2\left(1+\frac{g_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)\right]^{-2} \\ &\le G(g_{ms}^{\omega}, \rho_m^{\omega}) - \frac{1}{2}(c_1\cdot\tilde{g}_{ms}^{\omega})^2 - \frac{1}{2}\left[\tilde{\rho}_m^{\omega}\log_2\left(1+\frac{\tilde{g}_{ms}^{\omega}|h^{\text{Sat}}|^2}{\tilde{\rho}_m^{\omega}\cdot\sigma_0^2}\right)\right]^{-2} \\ &\qquad - c_1^2\cdot\tilde{t}_m^{\omega}\cdot(t_m^{\omega}-\tilde{t}_m^{\omega}) - \left[\tilde{\rho}_m^{\omega}\log_2\left(1+\frac{\tilde{g}_{ms}^{\omega}|h^{\text{Sat}}|^2}{\rho_m^{\omega}\cdot\sigma_0^2}\right)\right]^{-2}\cdot\begin{bmatrix}\frac{\partial H(\rho_m^{\omega}, g_{ms}^{\omega})}{\partial\rho_m^{\omega}} \\ \frac{\partial H(\rho_m^{\omega}, g_{ms}^{\omega})}{\partial g_m^{\omega}}\end{bmatrix}^T\cdot\begin{bmatrix}\rho_m^{\omega}-\tilde{\rho}_m^{\omega} \\ g_{ms}^{\omega}-\tilde{g}_{ms}^{\omega}\end{bmatrix} \\ &\triangleq \tilde{E}_{m\omega}^{c2}(\rho_m^{\omega}, g_{ms}^{\omega}; \tilde{\rho}_m^{\omega}, \tilde{g}_{ms}^{\omega}) \end{aligned} \tag{23}$$

Algorithm 2: Solving for Communication Resources $\{\varrho, \mathcal{G}\}$ Between Gateway Buoys and Satellite.

Input: Optimization Problem P_2^{Sub}.
Output: $\{\varrho, \mathcal{G}\} = \{\rho_m^{\omega}, g_{ms}^{\omega}\}_{m\in\mathcal{N}^M, \omega\in\Omega}$
Initialize $\{\{\rho_m^{\omega(0)}, g_{ms}^{\omega(0)}\}_{\omega\in\Omega}, \alpha_m^{(0)}, \beta_m^{(0)}\}_{m\in\mathcal{N}^M}$.
while (1) **do**
  for $m = \{1, 2, \cdots, M\}$ **do**
    Calculate the Lagrangian function based on (25).
    Update decision variable $\rho_m^{\omega(\kappa+1)}$ based on (26).
    Update decision variable $g_m^{\omega(\kappa+1)}$ based on (27).
    Update Lagrange multipliers $\alpha_m^{(\kappa+1)}$, $\beta_m^{(\kappa+1)}$ based on (29).
  end for
  if $\max\limits_{m,\omega}\{\rho_m^{\omega(\kappa+1)} - \rho_m^{\omega(\kappa)}\} < \varepsilon_\rho \wedge \max\limits_{m,\omega}\{g_m^{\omega(\kappa+1)} - g_m^{\omega(\kappa)}\} < \varepsilon_g \wedge \max\limits_{m}\{\alpha_m^{(\kappa+1)} - \alpha_m^{(\kappa)}, \beta_m^{(\kappa+1)} - \beta_m^{(\kappa)}, \eta_m^{(\kappa+1)} - \eta_m^{(\kappa)},\} < \varepsilon$ **then**
    BreakWhile
  end if
  Update $\kappa = \kappa + 1$.
  // Part(b): SCA-based Power Iteration
  Initialize $\{\tilde{\rho}_m^{\omega(\kappa,0)}, \tilde{g}_{ms}^{\omega(\kappa,0)}\}_{i\in\mathcal{N},j\in\mathcal{N}}$ variables.
  while (1) **do**
    Update $\{\tilde{\rho}_m^{\omega(\kappa,\tilde{\kappa}+)}, \tilde{g}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}\}_{i\in\mathcal{N},j\in\mathcal{N}}$ by solving the convex optimization problem (28) with substitute variable $\tilde{E}_{m\omega}^{c2}(\rho_m^{\omega}, g_{ms}^{\omega}; \tilde{\rho}_m^{\omega}, \tilde{g}_{ms}^{\omega})$ in (23).
    if $\{\tilde{\rho}_m^{\omega(\kappa,\tilde{\kappa}+)}, \tilde{g}_{ms}^{\omega(\kappa,\tilde{\kappa}+1)}\}_{i\in\mathcal{N},j\in\mathcal{N}}$ converge **then**
      Break While
    end if
    Update $\tilde{\kappa} = \tilde{\kappa} + 1$.
  end while
end while
return $\{\varrho, \mathcal{G}\}$

algorithm (top algorithm) refer to the URL https://github.com/JiaxuanXie19/OffshoreBuoysNetwork.

IV. Numerical Simulation and Platform Testing

Simulation results of the bilevel edge computing architecture and HSJDI algorithm are illustrated in this section, and the field tests are conducted in the actual marine environment of Lingao County, Hainan Province, China. The main parameters for the simulation are listed as follows. The number of gateway buoys and detection buoy are $M = 4$ and $Q = 20$. The buoy communication frequency is 2 GHz, which is the L band or S band. The satellite communication frequency is 30 GHz, which is Ka band. The buoys are distributed within a range of $100\times$ 100 km^2, and the satellite altitude is 2000 km. The initial power resource, computing resource of each buoy node is 20 W, 10 Mcycles. $G_0 = 50$ dBi. The initial value of I_q and ϕ_q is [4], [6] kbit and [0.8, 1.2] Mcycles, and these value will be adjusted appropriately in different simulations. The computing resource of remote cloud is 1000 Mcycles.

A. Numerical Simulation

Fig. 4(a) illustrates the position of 24 buoys including 20 detection buoys (blue round dots) and 4 gateway buoys (red square dots) within 100×100 km^2. Then, based on Algorithm 1, the connection relationship between detection buoy and gateway buoys (including relay transmission relations) can be obtained, as shown by the black dotted line in Fig. 4(a). Then, our HSJDI algorithm is run in such network topology graph.

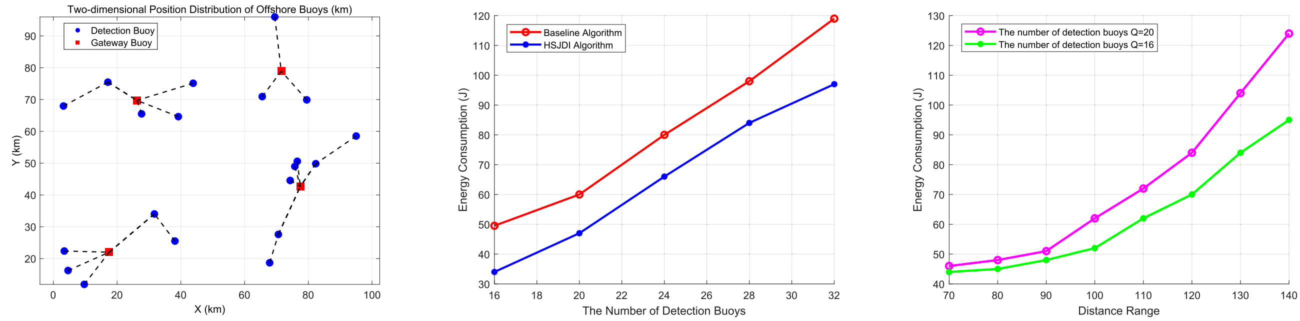
Fig. 4(b) compares our HSJDI algorithm and traditional marine networks algorithm as mentioned in [24] with the objective of minimizing the total energy consumption. However, this baseline algorithm jointly optimizes computation offloading, uploading transmission, and secrecy provisioning, ignoring the networking transmission among terminal devices (i.e., the networking transmission within buoys). Numerical simulations show that the performance of our HSJDI algorithm is about 16.67%–45.59% better than that of the traditional baseline algorithm, which proves the effectiveness of the proposed algorithm.

Fig. 4(c) illustrates the relationship between the total energy consumption and communication distance. Specifically, we adjust the communication distance range from $100\times$ 100 km^2 to a minimum of $70\times$ 70 km^2 and a maximum of $140\times$ 140 km^2. Numerical simulations reveal that the communication distance of the buoy plays a pivotal role in system performance. As such, it is recommended that the deployment range of buoys be carefully constrained to avoid excessive distances.

B. Platform Testing

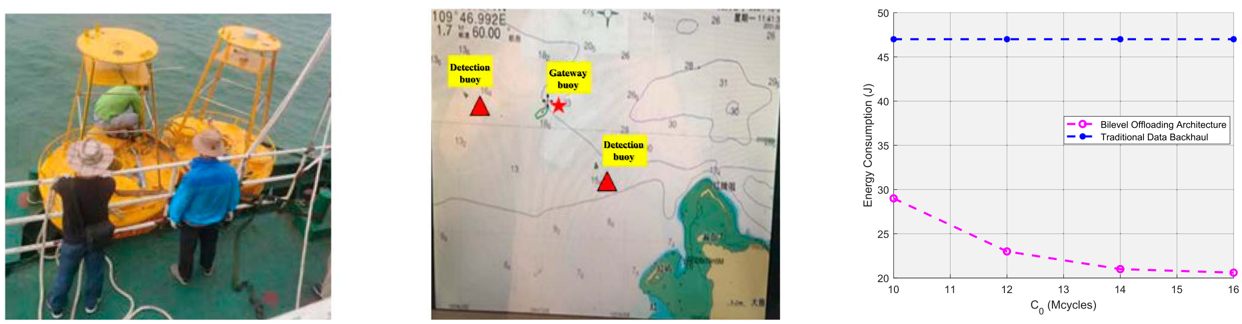
Finally, Fig. 5(a) and (b) illustrates the deployment of two detection buoys and one gateway buoy for platform testing with adequate energy, which aims to verify our bilevel edge computing architecture as well as our HSJDI algorithm. Specifically, we compare the performance of traditional data feedback and our bilevel offloading architecture by hardware platform, change some parameters, and set the CPU resources on the buoy node as 10–16 Mcycles. Besides, in our real maritime buoy system, energy consumption is measured through modular power aggregation and real-time monitoring during operational periods. The total energy consumption is calculated by summing the power indicators of all active modules (e.g., sensors, communication units, and computing units) within the buoy. Each module is equipped with subcircuit current sensors to record instantaneous power usage, and finally, the energy consumption is obtained from the calculation formula above.

The test results show that our bilevel offloading architecture can reduce the energy consumption by about 30.95%–50.95%, which verifies the effectiveness of our proposed architecture and offloading algorithm. The 30.95%–50.95% energy reduction range was derived by statistically analyzing the energy consumption of gateway buoys under varying CPU resource allocations while maintaining identical computational and communication tasks. Specifically, we quantified the energy consumption while systematically adjusting the total available CPU resources (e.g., clock frequency and core utilization) and measured the resultant energy expenditure. The observed range reflects the achievable efficiency gains across diverse hardware configurations, validating



(a) (b) (c)

Fig. 4. Simulation Results. (a) illustrates the position and the connection results within marine buoys. (b) compares the performance of HSJDI algorithm and baseline algorithm without networking transmission within buoys. (c) analyzes the impact of transmission distance on the total delay.



(a) (b) (c)

Fig. 5. Buoy platform testing. (a) and (b) illustrate the deployment of two detection buoys and one gateway buoy for platform testing. (c) is the performance comparison between the traditional data backhaul and the bilevel offloading architecture.

the adaptability and effectiveness of our approach. In addition, when the parameter C_0 changes, the total energy consumption changes monotonically with no significant fluctuations or anomalies, which implies that our bilevel architecture can guarantee robustness. At last, it is worth mentioning that maritime channel conditions are worse than expected. If real real-time channel statistics can be obtained, the accuracy of model calculation can be further improved.

V. Conclusion

Offshore buoys are essential for oceanic data collection and marine monitoring, but their limited capacity and heterogeneous satellite systems challenge data transmission. A bilevel edge computing architecture is proposed. Detection buoys offload tasks to gateway buoys or the cloud, leveraging their resources to ease communication bottlenecks. We address the NP-hard collaborative offloading problem by introducing the HSJDI algorithm, which simplifies computation through decomposition. Simulations confirm HSJDI's superiority over baseline methods, effectively reducing energy consumption. The field tests validate the platform and architecture.

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Jiaxuan Xie (Member, IEEE) received the Ph.D. degree in electrical engineering from Tsinghua University, Beijing, China, in 2025.

She has published a monograph and more than 30 papers and patents on communication. Her research interests include marine satellite Internet of Things and air-sea cross-domain communication.

Linling Kuang (Member, IEEE) received the B.S. and M.S. degrees from the National University of Defense Technology, Changsha, China, in 1995 and 1998, respectively, and the Ph.D. degree in electronic engineering from Tsinghua University, Beijing, China, in 2004.

Since 2007, she has been with Tsinghua Space Center, Tsinghua University. Her research interests include wireless broadband communications, signal processing, and satellite communications.

Dr. Kuang is a member of the IEEE Communications Society.

Honglei Mu (Member, IEEE) received the master's degree in computer science from Beihang University. He is currently working toward the Ph.D. degree in cybersecurity with Tsinghua University, Beijing, China.

Currently, he serves as the Chairman of Zhongguancun Robotics Industry Innovation and Development Company Ltd., Beijing, China. His research interests include unmanned systems and intelligent equipment such as unmanned aerial vehicles (UAVs), loitering munitions, unmanned tanks, quadruped robots, composite robots, explosion-proof inspection robots, and intelligent inspection equipment. He is also engaged in the research and development of a highly autonomous unmanned vehicle-mounted loitering intelligent platform. Based on these projects, he has published four academic papers and holds multiple patent certifications.